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SECTION 7- MAGNET RAMPING

WARNING!

THE FOLLOWING REQUIRED SAFETY ACTIONS MUST BE TAKEN PRIOR TO RAMPING THE MAGNET:

1. MAKE SURE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON, OR THE HATCH IS OPENED IF A MOBILE VAN, BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. REVIEW AND FOLLOW CRYOGEN SAFETY MEASURES CONTAINED IN SECTION 5-3 OF THE INTRODUCTION (CRYOGEN SAFETY).
2. NOTIFY SITE ADMINISTRATION BEFORE RAMPING THE MAGNET THAT ALL MAGNETIC SAFETY PRECAUTIONS MUST BE TAKEN.
3. POST WARNING SIGNS OUTSIDE THE 5 GAUSS ZONE TO ALERT PERSONNEL WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS AND OTHER BIOSTIMULATION DEVICES NOT TO PROCEED INTO THE DESIGNATED AREA. POST THESE SIGNS ON THE MAGNET ROOM LEVEL AS WELL AS AREAS BELOW THE MAGNET TO WHICH THE 5 GAUSS ZONE EXTENDS. SEE INTRODUCTION, SECTION 5.
4. POST "RAMPED MAGNET" WARNING SIGN AT THE MAGNET ROOM ENTRANCE PRIOR TO RAMPING THE MAGNET. WARNING TO ALERT PERSONNEL THAT NO FERROMAGNETIC MATERIAL OR INDIVIDUALS WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS OR STEEL PLATES ARE ALLOWED IN THE MAGNET ROOM WHEN THE MAGNET IS RAMPED.
5. REMOVE ALL LOOSE FERROMAGNETIC MATERIAL FROM THE MAGNET ROOM. PULL THE POWER SUPPLIES AS FAR AWAY FROM THE MAGNET AS THE CABLES AND SITE GEOMETRY ALLOW. METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.
6. MAKE SURE THAT THE EMERGENCY/ MAGNET RUNDOWN UNIT IS INSTALLED AND OPERATIONAL TO ENABLE THE MAGNETIC FIELD TO BE QUICKLY DISCHARGED IN CASE OF AN EMERGENCY. SEE SET UP AND CALIBRATION, SECTION 1-4 AND 1-6.
7. MAKE SURE THAT THE MAGNET IS AT LEAST 75% FULL OF LIQUID HELIUM TO PREVENT THE LIQUID HELIUM LEVEL FROM DROPPING TO A POINT, DURING RAMPING, WHERE A QUENCH MAY OCCUR.

7.1 PREPARATION

1. Set up the field monitoring equipment Probe and Teslameter in accordance with Set Up and Calibration, Section 5.
2. Make sure that the Magnet Power Supply is installed, checked and adjusted in accordance with the Vendor Manual (Sections 1 through 4) supplied with the unit. Verify Main Switch Heater Power Supply set for 815 $\pm$ 5 mA.

7-1 PREPARATION (continued)

3. Make sure that the Input Power Cable for the Power Supply is disconnected.
4. Connect the Power Supply to the magnet by making all cable connections in accordance with Set Up and Calibration, Section 6.
5. Record the Main Coil connection polarity In Functional Checks, Table 8-4.
6. Set all heater switches to the "OFF" position. Set "CURRENT CONTROL" and "VOLTAGE CONTROL" to 0 (full CCw).
7. Connect the Input Power Cable for the Main Power Supply.

7-2 RESISTANCE CHECKS

1. Check Switch Heater resistances in accordance with Functional Checks, Section 2.
2. Make sure "CURRENT CONTROL" and "VOLTAGE CONTROL" on the Main Power Supply are off (full ccw). Turn on Power Supply Input Power.
3. Turn on Axial Shim Heaters and observe current rise in ammeter (800–820 mA) to ensure circuit continuity, Make sure Main Heater Switch is off.

WARNING!

AXIAL SHIM SWITCH HEATERS MUST REMAIN ON DURING THE ENTIRE RAMPING PROCESS TO PREVENT IRREPARABLE SHIM COIL DAMAGE AND MAGNET QUENCH DURING RAMPING. THE POWER SUPPLY WILL NOT PASS CURRENT IN THE MAIN POWER LEAD CIRCUIT WITH THE AXIAL SHIM HEATERS OFF.

4. Connect a Digital Volt Meter (DVM) to the end of the Voltage Sense Leads terminated at the power supply.
5. Set "VOLTAGE CONTROL" on power supply to maximum (full CW).
6. Observe the Main Power Supply Ammeter and slowly turn the "CURRENT CONTROL" (CW) to set 500A current through the Main Power Leads, Lead Extensions and persistent Main Switch.
7. Record the voltage reading on the (DVM)_____ mV.

7-2 RESISTANCE CHECKS (continued)

8. A voltage reading less than 150 mV at 500 Amps indicates acceptable internal contact resistance of the Lead Extensions. Higher resistances will add more heat to the magnet increasing boil-off and possibly causing a quench during ramping.

If the voltage reading is greater than 150 mV, then wait approximately minute with the current running, reading may drop as the leads cool. If this fails, tighten the plate nuts on the Ramp Lead Hold Down Tool to move the leads slightly. Observe the effect on the voltage reading.

if the reading still exceeds 150 mV, then turn the "CURRENT CC W), turn off input power, and check/tighten the bolts securing Cables to the Power Supply and Ramp Leads. Lift and re-seat the Ramp Leads Repeat Steps 1 - 7.

Note

Repeated failing of the contact resistance check indicates a need to replace the lead Contact Bands. See Replacment/Maintenance, Section 11, or damaged Ramp Leads.

Upon passing the internal resistance check, continue with Step 9.

9. Set the Power Supply Voltmeter to read voltage across the Power Supply Output Lugs (read on Power Supply Voltmeter with toggle switch in Main Power Supply position).
10. Gradually increase the "CURRENT CONTROL" (CW) to pass 735A through the IMain Power I-cads, I-cad Extensions and persistent Main Switch while observing the Power Supply Voltmeter. If the voltage exceeds 2.2V, discontinue the test.

Note

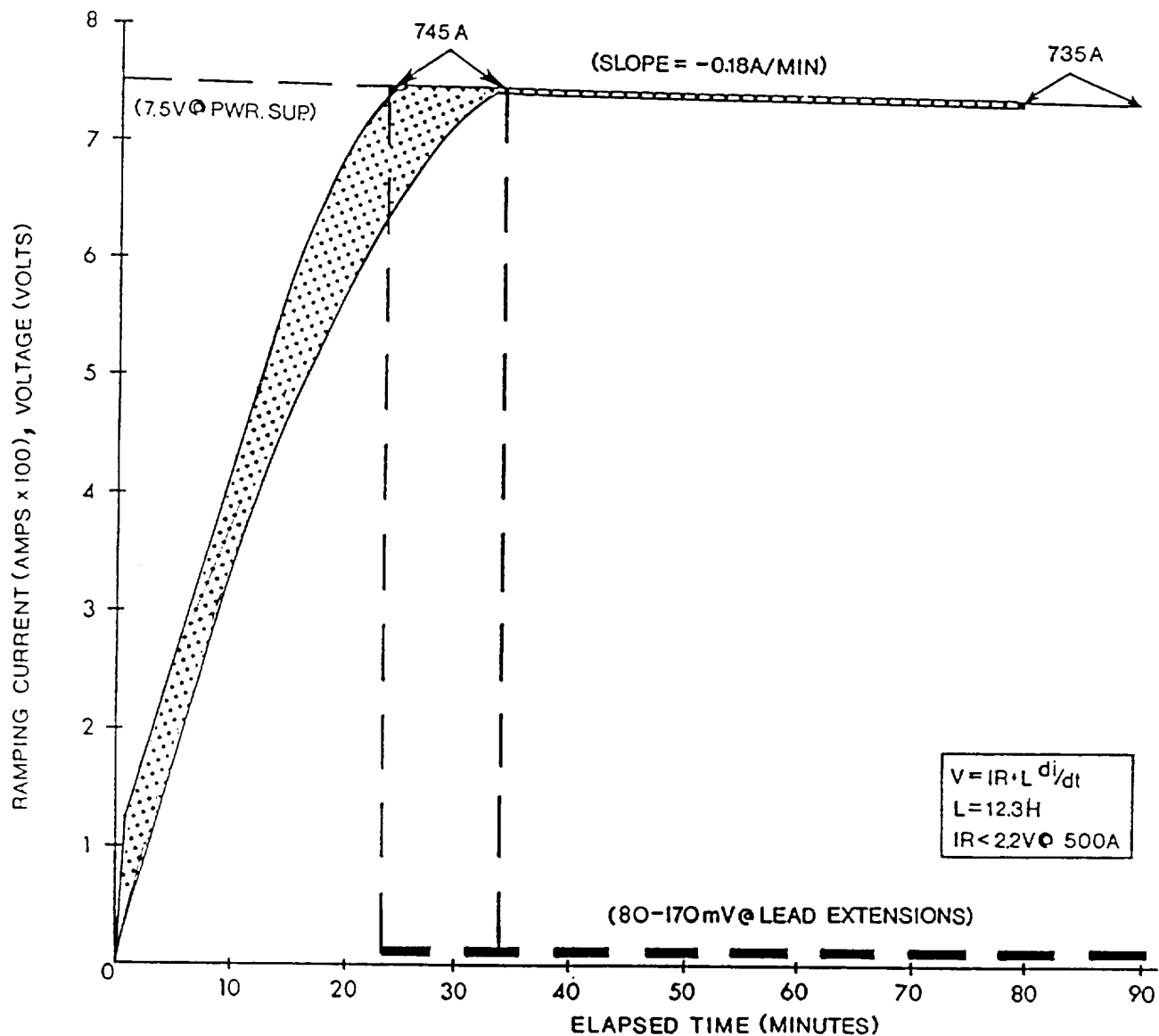
A voltage less than 2.2V at 735A indicates acceptable system resistance. If the voltage exceeds 2.2V during the test, follow the procedures in Step 8 for adjusting contact resistance.

11. Upon passing Step 10, turn the "CURRENT CONTROL" and "VOLTAGE CONTROL" off (full CCW) and continue with the ramping procedure.

7-3 RAMPING

Description:

The Ramping Procedure covered here provides for a rapid ramping slightly above field with a gradual reduction to field (1 .5 T). This method has resulted in the fastest "settling" time. See Illustration 7-1.



MAGNET RAMPING PROFILE
ILLUSTRATION 7-1

7-3 RAMPING (continued)**Procedure:**

If a Quench occurs during ramping, immediately turn "VOLTAGE CONTROL" and "CURRENT CONTROL" of the Main Coil Power Supply to zero (fully CCW). A quench is a rapid discharge of the magnetic field which will result in the rapid generation and expulsion of helium gas, rupturing the Burst Disc In the Vent System.



MAKE SURE THAT THE AXIAL SHIM SWITCH HEATERS ARE ON DURING RAMPING TO PREVENT COIL DAMAGE AND MAGNET QUENCH. GE POWER SUPPLIES HAVE PROTECTIVE CIRCUITRY TO PREVENT RAMPING VOLTAGE WITH THE AXIAL HEATER SWITCH IN THE "OFF" POSITION.

1. Make sure Valve (V2) is closed.
2. Make sure "VOLTAGE CONTROL" and "CURRENT CONTROL" is at zero (full CCW).
3. Engage "SET POINT" button (depress and hold until current set) and set "CURRENT CONTROL" for a reading of 745A. This sets the maximum current output of the Power Supply.
4. Turn on Main Switch Heater, leave Axial Shim Switch Heater on. Wait 3 minutes.
5. Check that Main Switch Heater current is 815 ± 5 mA.
6. Turn "VOLTAGE CONTROL" on Main Power Supply to maximum (full CW).
7. Estimate the system inductance by measuring current change over a 10 second ramping interval:

$$L \text{ (inductance) } = 10 \times \text{Voltage/ Current Change}$$

Note

This method will give inaccurate values of inductance when the current is less than 200 Amps.

7-3 RAMPING (continued)

Note

Measured inductance should be approximately 12.3H (henry), if the calculated value is between 11 – 13.5H continue with the procedure. If the calculated value is outside this range, discontinue ramping and “troubleshoot” the system. Measure Main Coil Resistance (see Functional Checks, Section 2) and contact the Region Magnet Support Engineer.

8. Observe the voltage profile across the Magnet Voltage Taps (Power Supply Voltage Switch set to Main Coil) as the ramping current approaches 745A.
9. When current reaches 745A, start decreasing current by turning “CURRENT CONTROL” (CCW).
10. Decrease current at a rate of approximately 0.18 amp/min. to 735A and allow to stabilize.

Note

Voltage will go negative as current decreases.

11. Check Teslameter and adjust current as required, to bring magnetic field to within (63.87– 63.88 MHz) (15000.5 15002.0 gauss) range. Allow final current to stabilize. Voltage should read less than 30mV.
12. Maintain current at final setting until the Main Switch goes into the “persistent” mode in Step 13.
13. Turn off Main Switch Heater. Wait a minimum of 7 minutes for the switch to fully cool and go “persistent”.

Note

Observe voltage (read on Power Supply Voltmeter with toggle switch in Main Coil Position). When the switch goes “persistent” the voltage across the magnet terminals will drop to approximately 0.00V. This sudden voltage drop will indicate that the switch has gone into the persistent mode.

14. When the switch goes “persistent”, slowly turn the power supply “CURRENT CONTROL” to zero over a one minute period (Full CCW).

Note

Check that Teslameter does not decrease as current knob is turned. If it does, switch is not persistent.

15. Record current value at which the switch went “persistent” in Functional Checks, Table 8-4.

7-3 Ramping (continued)

WARNING!

ASSURE THAT THE CONNECTION POLARITY AND FINAL RAMPING CURRENT ARE RECORDED IN FUNCTIONAL CHECKS, TABLE 8-4. THIS INFORMATION IS ESSENTIAL FOR LATER CHANGING OF THE MAGNETIC FIELD. THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS TO AVOID A QUENCH WHEN TURNING ON THE MAIN SWITCH.

16. Gradually turn the "VOLTAGE CONTROL" to zero (over a one minute period) and turn the power supply off.
17. Turn off the Axial Shim Heaters.
18. Disconnect Input Power Cable from the Main Power Supply.
19. Slowly open valve (V2) while observing Pressure Gauge to repressurize the Cryostat. When pressure reaches 0.3 psig, close valve (V2).
20. Remove the Hold Down Tool. Disconnect the Main Power Leads, Volt Sense Leads and remove Main Power Lead Extensions on the top of the magnet. Immediately replace the caps onto the Main Power Lead Extension Receptacles.

CAUTION

Cryostat exhaust flow rates and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the Helium Exhaust System or Vent Valve (V2).

21. Open Vent Valve (V2) to de-pressurize the Cryostat to 0.25 psig. Close V2.

Note

Read all flow rates from the bottom of the float (ball) on the flowmeters.

Note

Flow rates may be temporarily elevated after ramping. Do not adjust them until after the magnet has had time to stabilize (at least one day).

7-3 Ramping (continued)

22. Set Flowmeter (F1) between 0.4 - 0.6 SCFH.
23. Set Flowmeter (F2) between 1.5 - 2.0 SCFH to maintain a Cryostat Pressure Gauge reading between 0.25 - 0.50 psig.
24. Ensure flow rate through F2 is equal or greater than 1.5 SCFH.
25. If flow rate through F2 is less than 1.5 SCFH or the pressure gauge reads less than 0.25 psig, pressurize the vessel and "bubble test" all exhaust plumbing joints, relief valve and Shim Lead Connector. Ensure V2 is fully closed. Repair any leaks. If a 1.5 SCFH flow rate through F2 cannot be achieved, under the above conditions, contact your Region Magnet Service Engineer.
26. Ensure the following conditions are maintained. Re-check settings in three days and again after one week:

FLOWMETER (F1) = 0.4 - 0.6 SCFH

FLOWMETER (F2) > 1.5 - 2.0 SCFH

CRYOSTAT GAUGE PRESSURE - 0.25 - 0.50 psig

27. Proceed to Set Up and Calibration, Section 8 (Shimming Preparation/Field Stabilization).